

Umut Varolgunes

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Employment

- Assistant Professor at Koç University: November 2023 - present
- Visiting Scholar at Boğaziçi University: November 2021 - November 2023
(under TUBITAK 2236 CoCirculation2; New Horizon 2020 Marie Skłodowska-Curie Actions Cofund Program, on leave Sep. 2021 - Feb. 2022 at University of Edinburgh)
- Szego Assistant Professor at Stanford University: September 2018 - June 2021

Education

Graduate - Massachusetts Institute of Technology

- September 2013 - May 2018
- PhD. in Mathematics
- Advisor: Paul Seidel
- Thesis title: Mayer-Vietoris property for relative symplectic cohomology

Undergraduate - Massachusetts Institute of Technology

- September 2009 - May 2013
- Bachelor of Science degrees in Mathematics and Physics

Research Interests

symplectic geometry, Floer theory, algebraic geometry

Awards and Honors

Gold Medals in 2008 and 2009 International Mathematics Olympiads
MIT Praecis Presidential Fellowship 2013 - 2014
BAGEP Award of the Science Academy 2024

Languages

Turkish (native), English (fluent)

Publications

Descent with algebraic structures for symplectic cohomology, Proceedings of the 29th Gokova Geometry/Topology conference (2025), 18-36

Framed E_2 structures in Floer theory (joint with Mohammed Abouzaid and Yoel Groman), Advances in Mathematics 450 (2024): 109755

A characterization of heaviness in terms of relative symplectic cohomology (joint with Cheuk Yu Mak and Yuhan Sun), Journal of Topology 17 (1), e12327

Locality of relative symplectic cohomology for complete embeddings (joint with Yoel Groman), Compositio Math. 159.12 (2023), 2551–2637

On homological mirror symmetry for chain type polynomials (joint with Alexander Polishchuk), Mathematische Annalen 388.3 (2024), 2331–2386

On certain quantifications of Gromov’s non-squeezing theorem (joint with Kevin Sackel, Antoine Song and Jonathan Zhu), Geometry & Topology 28.3 (2024): 1113–1152

Quantum cohomology as a deformation of symplectic cohomology (joint with Nick Sheridan and Strom Borman), JFPTA Viterbo 60 Volume 24.2 (2022): 1–77

Seifert form of chain type singularities, Kyoto Journal of Math 63.1 (2023): 155–193

Super-rigidity of certain skeleta using relative symplectic cohomology (joint with Dmitry Tonkonog), Journal of Topology and Analysis, Volume No. 15 (2023), Issue No. 01, 57–105

Mayer-Vietoris property for relative symplectic cohomology, Geometry & Topology 25.2 (2021), 547–642

On the equatorial Dehn twist of a Lagrangian nodal sphere, Proceedings of the 22nd Gokova Geometry/Topology conference (2017), 112–137

Preprints

Closed string mirrors of symplectic cluster manifolds (joint with Yoel Groman) - <https://arxiv.org/abs/2211.07523>

Contact big fiber theorems (joint with Yuhan Sun and Igor Uljarevic) - <https://arxiv.org/abs/2503.04277> - accepted for publication in Geometric And Functional Analysis

Grants awarded	2026-28	TÜBİTAK/JSPS	Fukaya Categories, Sheaves and Quantization of Cotangent Bundle	2.316.000 TRY
	2025-28	TÜBİTAK	Symplectic invariants of Calabi-Yau varieties	1.180.000 TRY
	2021-23	TÜBİTAK/EU cofunded	Non-archimedean analytic geometry and Lagrangian torus fibrations	143.520 €
Organizational Activity	Cofounder of Kylerec graduate student workshops			
	Organizer for Stanford Symplectic Geometry Seminar 2018-2020			
	Organizer for North California Symplectic Geometry Seminar 2018-2020			
	Organizer for the conference “Mirror symmetry for Looijenga interiors and beyond”			
	Organizer for the “Dual perspectives” seminars			
	Organizer for the workshop “Geometry in İstanbul”			
	Organizer for the workshop “Mirror symmetry and differential equations”			
	Organizer for Koç Math Seminars 2023-2024			
Students	Organizer for İstanbul Geometry/Topology Seminars 2024-2025 (joint with Bogazici University)			
	Berk Dinçay (Master, 2025-)			
	Cem Yalım Özel (PhD, 2025-)			

Teaching

TA for 18.01/02A (Expedited single and multi variable calculus), Fall 2015, MIT

TA for 18.03 (Differential equations), Spring 2016, MIT

TA for 18.095 (Winter break lecture series), Winter 2018

Lecturer for Math 113 (Linear algebra with proofs, undergraduate), Winter 2019, Stanford

Lecturer for Math 215B (Differential topology, graduate), Winter 2019, Stanford

Lecturer for Math 257C (Hamiltonian Floer theory, advanced graduate), Autumn 2019, Stanford

Lecturer for Math 257A (Symplectic geometry, graduate) Fall 2019, Stanford

Lecturer for Math 147 (Differential topology, undergraduate) Fall 2019, Stanford

Lecturer for Math 53* (Ordinary differential equations, undergraduate) Spring 2020, Stanford

Lecturer for Math 215B (Differential topology, graduate), Winter 2021, Stanford

Lecturer for Math 53* (Ordinary differential equations, undergraduate) Spring 2021, Stanford

Lecturer for Math 113 (Linear algebra with proofs, undergraduate), Spring 2021, Stanford

Mini course on relative symplectic cohomology (8 lectures) - Edinburgh, Fall 2021

Lecturer for Math 58J (Cohomology of manifolds I, undergraduate), Spring 2022, Boğaziçi University

Lecturer for Math 58T (Cohomology of manifolds II, undergraduate), Spring 2023, Boğaziçi University

Lecturer for Math 405 (Differential geometry, undergraduate), Spring 2024, Koç University

Lecturer for Math 456 (Cohomology of manifolds, graduate), Fall 2024, Koç University

Lecturer for Math 208 (Advanced Calculus, undergraduate), Spring 2025, Koç University

Lecturer for Math 402 (Topology, undergraduate), Spring 2025, Koç University

Lecturer for Math 203* (Multivariable Calculus, undergraduate), Fall 2025, Koç University

*more than 100 students

Invited talks

Preprint seminar on symplectic geometry and related topics - Columbia University, October 12, 2015

23rd Gokova Geometry and Topology Conference - May 30 - June 4, 2016

Symplectic Geometry Seminar - Stanford University, January 30, 2017

Student Symplectic Seminar - Stanford University, February 9, 2017

Math-Physics Joint Seminar - UPenn, April 18, 2017

Symplectix seminar - Paris, September 15, 2017

Geometry/Physics seminar - Northwestern University, September 28, 2017

Topology and Symplectic Geometry / Math of Gauge Fields seminar - Stony Brook University, October 5, 2017

Symplectic Geometry Seminar - Stanford University, October 9, 2017

Seminar on Mirror Symmetry - UC Berkeley, October 13, 2017

Topology Seminar - Princeton University, October 19, 2017

Mini course on my thesis (4 lectures) - Columbia University, Spring 2018

Geometry/Topology Seminar - UC Davis, January 22, 2019

North Carolina Symplectic Geometry Seminar - Stanford University, May 6, 2019

Mathematics seminar - Korean Institute of Advanced Study, Seoul, August 30, 2019

Symplectic seminar - Stony Brook University, September 12, 2019

Symplectic geometry, Gauge theory and Categorification Seminar - Columbia University, September 13, 2019

Simons Colloboration on Homological Mirror Symmetry Annual meeting - New York, November 14, 2019

Topology Seminar - University of Oregon, March 10, 2020

Geometry, Symmetry and Physics Seminar - Rutgers University (online), May 14, 2020

M-Seminar - Kansas State University (online), May 28, 2020

Western Hemisphere Symplectic Geometry Seminar (online) - June 26, 2020

UCGEN Seminar (online) - September 23, 2020

Montreal, Princeton/IAS, Paris, Tel-Aviv Symplectic Zoominar (online) - October 16, 2020

Boston University Geometry & Physics seminar (online) - March 24, 2021

Beijing-Novosibirsk seminar on geometry and mathematical physics (online) - April 15, 2021

Shafarevich Seminar - Steklov Institute, September 21, 2021

3CinG Annual Congress (online) - September 22, 2021

Friday seminar (?) - Higher School of Economics, September 24, 2021

Freemath (online) - September 28, 2021

Differential geometry and topology seminar - Cambridge University, October 27, 2021

Hodge Seminar - Bayes Center, Edinburgh, November 4, 2021

Geometry and dynamics seminar - Tel-Aviv University (online), November 10, 2021

Symplectic and Poisson geometry seminar - University of Illinois at Urbana-Champaign (online), November 29, 2021

Geometry and Topology seminar - Michigan State University (online), December 7, 2021

Workshop on Mirror symmetry and Related Topics - Kyoto (online), December 13, 2021

Global Youth Forum - USTC-IGP, Hefei (online), January 11, 2022

Nick Sheridan's postdoc seminar - Bayes Center, Edinburgh, January 19, 2022

GeNeSys: Geneva-Neuchâtel Symplectic geometry seminar, January 24, 2022

Montreal, Princeton/IAS, Paris, Tel-Aviv Symplectic Zoominar (online) - February 18, 2022

IMBM - Istanbul, February 17-22-25, 2022

Geometry and dynamics seminar (TIDY Distinguished Speaker) - Tel-Aviv University, April 27, 2022

Pure Maths Colloquium - Sheffield University (online), May 25, 2022

Math Department Colloquium - Koc University, October 20, 2022

Math Seminar - Bilkent University, November 16, 2022

Symplectic Geometry, Gauge Theory, and Categorification Seminar - Columbia University, December 16, 2022

Topology Seminar - University of Southern California, January 11, 2023

Symplectic geometry seminar - Stanford University, January 19, 2023

Symplectic Geometry, Gauge Theory, and Categorification Seminar - Columbia Uni-

versity, January 31, 2023

Geometria em Lisboa - IST (online), March 14, 2023

Hong Kong University of Science and Technology - Mirror symmetry via relative invariants, March-April, 2023 (6 hours)

M-Seminar - Kansas State University (online), April 7, 2023

Edge Seminar - Bayes Center, Edinburgh, May 17, 2023

The London Geometry and Topology Seminar - Imperial College, London, May 19, 2023

28th Gokova Geometry and Topology Conference, May 29 - June 3, 2023

29th Nordic Congress of Mathematicians (with EMS) at Aalborg University, July 3-7, 2023

Workshop on Symplectic Topology - University of Belgrade, August 21 - 25, 2023

GeoStaR 2 Workshop - Belalp, Switzerland, September 17-19, 2023

UMass Boston Math Colloquium, October 24, 2023

General Math Seminar - Gebze Technical University, April 5, 2024

Bonn Symplectic Geometry Seminar - Max Planck Institute, May 17, 2024

Toric and Tropical Techniques in Symplectic Field Theory - Mittag-Leffler Institute, Sweden, June 24-28, 2024

MIST Workshop - Chinese University of Hong Kong, August 12-20, 2024 (6 hours)

Seminar on Field Theory and String Theory - Shanghai Institute of Mathematics and Interdisciplinary Sciences, August 29, 2024

International Conference on Differential Geometry - Istanbul University, September 2-5, 2024

IMBM Mathematics Days VIII - October 18, 2024

Symplectic Geometry Seminar - Stanford University, November 15, 2024

Recent Advances in Mirror Symmetry and Degenerations - University of Miami, November 18-22, 2024

Workshop on Mirror symmetry and Related Topics - Kyoto, December 9-13, 2024

Symplectic Geometry Seminar, ETH - Zurich, March 31, 2025

Balkan Mathematics Conference II, Aristotle University of Thessaloniki - Thessaloniki, June 12-14, 2025

Celebrating the Consortium IMSAC, International Centre for Mathematical Sciences - Sofia, August 7-9, 2025

SIMIS - Symplectic invariants with support and mirror symmetry (mini course, 12 hours) - Shanghai, September 2025

Homological Mirror Symmetry 2026, IMSA - Miami, January 29 - February 2, 2026 (3 lecture mini course)